Toward Symbolic-Aware Music Language Models

Compact Learned Symbolic Planning Tokens for Controlled Music Fusion

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Audio language models handle sound well — but musical structure lives in MIDI, not in waveforms. **Can we inject symbolic structure into a LALM in a controlled, verifiable way?**

Audio Tokens Do Not Reliably Encode Musical Structure

What audio tokens can encode	What they miss
Timbre, loudness, texture	Key signature, chord progression
General instrument colour	Stem-level instrument identity
Rough tempo feel	Exact meter and beat phase
High-level scene	Phrase boundaries, section structure

MIDI encodes these exactly: note events, instrument tracks, tempo and key events. We learn a *compact* representation of this information and inject it into a LALM.

Verifying that symbolic injection is *useful* requires more than comparing to no-injection: a shuffled or dummy stream of the same length could produce the same gain from capacity alone. **Our whole evaluation is designed around this distinction.**

Three Research Questions Organize This Work

Question	What we need to show
RQ1	
Acoustic utility	Aligned symbolic stream reduces acoustic token cross-entropy more than a shuffled or dummy stream of the same size.
RQ2	
Alignment specificity	The gain comes from *alignment-specific content*, not token budget or model capacity. Requires aligned $>$ shuffled AND aligned $>$ dummy.
RQ3	
Structural content	Symbolic stream encodes music-structural attributes recoverable by probing — verified against a zero-feature baseline that exposes dataset priors.

Experiments are **evidence for these claims**, not the claims themselves. Each comparison uses the same aligned / shuffled / dummy protocol so gains can be attributed precisely to symbolic content — not to extra tokens.

Our Approach: S_{plan} as a Third Stream in UniAudio 2.0

R_a **reasoning** → S_{plan} **symbolic** → C_a **acoustic**

R_a reasoning	S_{plan} symbolic	C_a acoustic
8 streams	4 streams	8 streams
151 frames	30 frames	376 frames

S_{plan} uses a 4-stage RVQ codec trained on MIDI-derived frame features (pitch-class, instrument, meter, density). Total sequence: $151 + 30 + 376 =$ **557** tokens — **0% truncation** under 1,024 limit.

Raw MIDI appending fails: 2,000–8,000 tokens per window saturates the context, and variable-length sequences cannot be shuffled or dummy-controlled uniformly. S_{plan}'s fixed-length RVQ design makes the evaluation protocol possible.

Evaluation Protocol: Every Comparison Uses All Four Conditions

A gain of aligned over both shuffled and dummy is required for an alignment-specific claim

Condition	Content of S_{plan} slot	What a gain here proves
Aligned S_{plan}	Tokens from the *same* window's MIDI	Symbolic content + temporal alignment
Shuffled S_{plan}	Tokens from a *different* window's MIDI	Token statistics intact; alignment broken
Dummy stream	Zero / random RVQ codes	Token-budget / capacity effect only
Reason-only	No S_{plan} slot at all	Base model without any symbolic stream

Claim rule: aligned must beat *both* shuffled and dummy. Shuffled preserves statistics but breaks alignment; dummy removes all content. Together they isolate the alignment-specific effect from pure capacity.

Prerequisite: The Symbolic Codec Reconstructs Faithfully

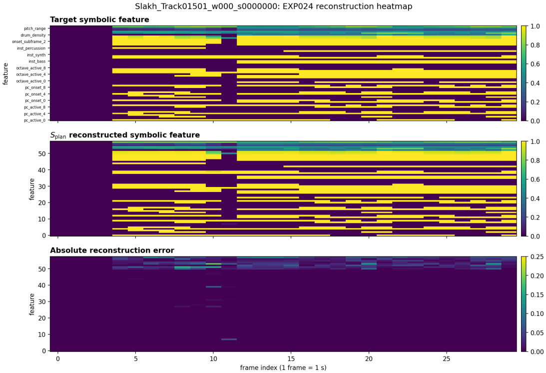
4-stage RVQ, 512 codes per stage — 42,000 training steps on Slakh

Metric	Value
Validation reconstruction loss	0.00918
Binary feature accuracy	**0.99777**
Continuous feature MSE	**0.00172**
Codebook utilization — books 1/2/3/4	31% / 72% / 81% / 81%
Fixed window `w000` binary acc / MSE	1.000 / 0.00174
Fixed window `w001` binary acc / MSE	1.000 / 0.00246
Fixed window `w003` binary acc / MSE	1.000 / 0.00130

Books 2–4 reach 72–81% utilization — no codebook collapse. This is a **prerequisite**, not a claim: it establishes that differences in downstream comparisons come from symbolic content, not codec failure.

Prerequisite: Feature Heatmap — Target vs. Reconstruction

Slakh Track01501, window 0 (0–30 s) — binary acc 1.000, MSE 0.00174



Rows: feature dimensions (binary instrument flags, pitch-class bits, continuous values). Columns: 30 time frames (1 s each). Left = ground-truth MIDI features. Right = S_{plan} reconstruction.

RQ1 — First Positive Signal: Proxy Prediction Task

Frozen codec + small transformer head predicting acoustic tokens from symbolic representations

Condition	Valid CE	Test CE	Test PPL
Reason only	7.565	7.657	2116
Low-rate rule summary	7.553	7.645	2089
Aligned S_{plan}	**7.507**	**7.596**	**1991**
$\hookrightarrow$ Shuffled S_{plan}	7.538	7.631	2061
$\hookrightarrow$ Dummy stream	7.602	7.694	2196
$\hookrightarrow$ Random stream	7.540	7.632	2064

Aligned beats reason-only ($\Delta = -0.061$) and shuffled ($\Delta = -0.034$). Learned S_{plan} outperforms the hand-crafted rule summary (-0.048 vs. -0.013).

First evidence for RQ1. Aligned beats both corrupted controls. Temporal alignment of symbolic content to its own audio window drives the gain. Claim boundary: proxy task; end-to-end integration is tested separately.

RQ1 — Replication: Stream-Native Interface Confirms the Gain

Full $R_a \to S_{\mathrm{plan}} \to C_a$ layout with matched positional encoding

Condition	Test CE	Test Acc	Gate
Aligned S_{plan}	**5.766**	**0.1097**	0.003
$\hookrightarrow$ Shuffled	5.785	0.1067	0.003
$\hookrightarrow$ Dummy	5.785	0.1068	0.003

Aligned beats shuffled ($\Delta = -0.018$) and dummy ($\Delta = -0.019$) consistently. Sequence 557 tokens, **0% truncation**.

RQ1 holds with end-to-end integration. The positive gap replicates when S_{plan} is inserted as a native third stream — not just as an external side-channel. This rules out a proxy-task artifact.

RQ1 — Primary Result: Hardened Four-Condition Comparison

All four conditions identical except stream variant — matched hyperparameters, fixed seed

Condition	Test CE	Test PPL	Acc	Gate
Reason-only baseline	5.859	350	0.104	0.000
Aligned S_{plan}	**5.600**	**270**	**0.121**	**0.177**
$\hookrightarrow$ Shuffled S_{plan}	5.614	274	0.119	0.167
$\hookrightarrow$ Dummy stream	5.614	274	0.119	0.163

Aligned beats both corrupted controls ($\Delta = -0.014$) on top of a large capacity gain (-0.245 vs. reason-only).

RQ1 answered positively — most rigorous evidence. Gate confirms the model opens the symbolic slot more for aligned (0.177) than shuffled (0.167) or dummy (0.163).
Boundary: oracle MIDI input; semantic-token CE only.

RQ2: The Gain Is Alignment-Specific, Not Token Capacity

The −0.259 CE gain over reason-only decomposes into two distinct effects:

Effect	Δ CE
Stream capacity (any symbolic stream vs. reason-only)	−0.245
Alignment-specific (aligned vs. both controls)	−0.014
Total (aligned vs. reason-only)	**−0.259**

The alignment-specific effect is consistent across all three controlled comparisons:

Comparison	Setup	Aligned vs. shuffled
Proxy task	Frozen codec, 5,000 steps	−0.034
Stream-native warmup	End-to-end, 3,000 steps	−0.018
Hardened comparison	Matched, 800 steps	−0.014

RQ2 answered positively. Consistent direction across independent setups rules out a single-

RQ3: Music Structure Is Recoverable from S_{plan}

Supervised probing on frozen embeddings vs. zero-feature baseline (head bias only)

MIR attribute	Aligned S_{plan}	Zero-feature control
Weak key accuracy	**0.557**	0.126
Pitch-class macro-F1	**0.854**	0.830
Instrument macro-F1	**0.501**	0.426
Segment density acc.	0.874	0.879
Instrument micro-F1	0.791	0.835
Meter accuracy	0.783	0.855

Red = zero-feature control outperforms — these axes are prior-dominated.

High accuracy alone is *not* evidence — a zero-feature baseline is mandatory. Weak key (+0.431 gap) and pitch-class (+0.024 gap) are the strongest safe axes. **RQ3: qualified positive answer** — key and pitch-class confirmed, meter and instrument micro-F1 excluded.

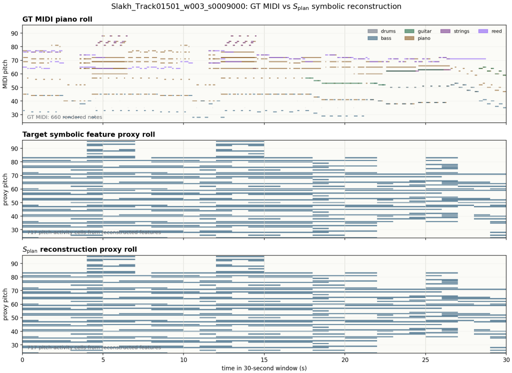
RQ3: Which Axes Can We Claim?

Organized by what the zero-feature control reveals about each attribute

Axis	Gap vs. zero-feat.	Status
Weak key accuracy	$+0.431$	**Safe to claim.** Label-shuffle control pending.
Pitch-class macro-F1	$+0.024$	Positive but small; target-prior control needed.
Instrument macro-F1	$+0.075$	Some signal; class imbalance must be resolved.
Instrument micro-F1	-0.044	Excluded. Common-class prior dominates.
Meter accuracy	-0.072	Excluded. Common-meter prior dominates

RQ3: Qualitative — Reconstruction vs. Ground-Truth Piano Roll

Slakh Track01501, window 3 (90–120 s) — binary acc 1.000, MSE 0.00130



Left: ground-truth MIDI piano roll. Right: pitch-region reconstruction from the S_{plan} codec.

Boundary: symbolic feature reconstruction — not MIDI transcription or waveform generation.

Design Constraint: Stream-Native Interface Is Necessary

Two negative results that establish the interface requirement — not incidental failures

Side-channel insertion:

Condition	Test CE	vs. reason-only
Reason-only	**5.360**	baseline
Aligned S_{plan}	5.511	$+0.151$
Shuffled S_{plan}	5.512	$+0.152$
Dummy	5.510	$+0.150$

Aligned $\approx$ shuffled $\approx$ dummy: model ignores symbolic content entirely. **Adapter gating:** gate converges to 3.4×10^{-4} — training stable, fusion absent.

Same root cause in both cases: symbolic tokens must enter through the same stream-native positional encoding as R_a and C_a. **This is why the RQ1/RQ2 positive results are non-trivial.**

What We Cannot Claim Yet

Gap	Why it matters
No decoded generation	CE results are on semantic tokens only. No waveform decoded from aligned vs. shuffled.
Oracle MIDI input	Aligned MIDI unavailable at inference on real audio. Needs audio-to-S_{plan} predictor.
No LLM-supervised tokenizer	Text-language grounding (open QA, captioning) not yet trained.
Pending MIR controls	Label-shuffle and target-prior controls for probing are incomplete.
Slakh only	No real-recording or cross-domain transfer evidence.

Next priority: decode C_a from the aligned checkpoint to waveform and compare perceptually against shuffled — converting the token-level claim into perceptual audio evidence.

Conclusion: Three Research Questions Answered

RQ	**Finding**
RQ1 Acoustic utility	**Positive.** Aligned S_{plan} reduces acoustic token CE in three independent controlled comparisons. Primary result: test CE 5.600 vs. reason-only 5.859.
RQ2 Alignment specificity	**Positive.** Aligned beats both shuffled and dummy consistently ($\Delta \in [-0.034, -0.014]$ across all three comparisons). Corroborated by gate values: 0.177 > 0.167 > 0.163.
RQ3 Structural content	**Qualified positive.** Weak key ($+0.431$) and pitch-class ($+0.024$) confirmed above the zero-feature baseline. Meter and instrument micro-F1 excluded as prior dominated.

Backup: Per-Codebook Cross-Entropy

Symbolic benefit distributed across all 8 acoustic codebooks (hardened comparison)

Condition	cb0	cb1	cb2	cb3	cb4	cb5	cb6	cb7
Reason-only	5.60	6.00	3.45	5.28	6.02	6.48	6.87	7.16
Aligned S_{plan}	**5.24**	**5.71**	**3.19**	**4.98**	**5.75**	**6.25**	**6.68**	**7.01**
Shuffled	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02
Dummy	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02

Symbolic benefit is distributed across all 8 codebooks. Books 0, 2, and 3 show the largest gains, suggesting symbolic structure acts as a long-range prior that helps coarser-grained acoustic prediction first.

Backup: Per-Window Case Spectrum (Stream-Native Warmup)

Both positive and boundary cases — no cherry-picking

Window	Aligned CE	Best control	Margin	Note
w000	5.226	5.280	$+0.054$	Strong positive
w001	6.159	6.163	$+0.004$	Small positive
w002	6.097	6.091	-0.006	Boundary / failure
w003	5.876	5.869	-0.007	Boundary / failure

Aggregate positive result holds even though individual windows show mixed margins. Reporting boundary cases prevents cherry-picking in qualitative reports.

Backup: Dataset — Slakh2100

Paired audio and MIDI, track-level split

Split	Tracks	Windows	Notes
Train	1,289	11,279	Track-level split before windowing
Validation	270	2,334	Zero track overlap with train
Test	151	1,385	Held-out; never seen during training

Track-level splitting prevents musical phrases from appearing in both train and test. All cross-entropy numbers in this presentation are on the held-out **test set** unless labelled “valid”.

All four conditions in every comparison use the same audio-MIDI windows and split. No experiment uses data outside Slakh.